

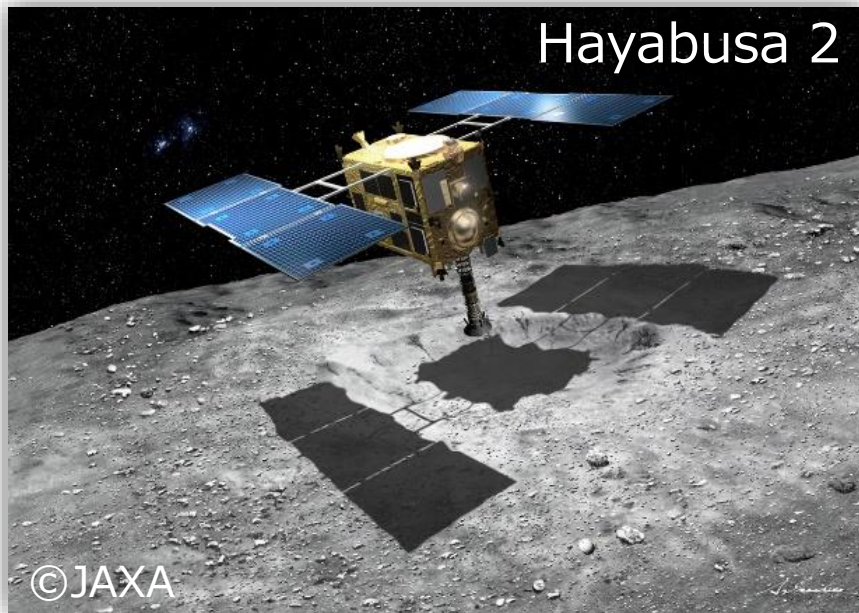


Transonic Free-Flight Experiment for the Dynamic Behaviour of a Next-Generation Deep-Space Sample Return Capsule

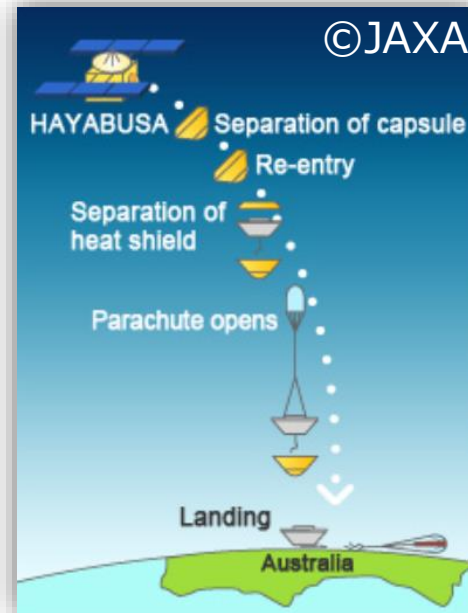
Finn Cracknell

(Tohoku University, Monash University)

■ Sample return mission



Hayabusa 2 Spacecraft



**Hayabusa Sample
Return Mission Profile**

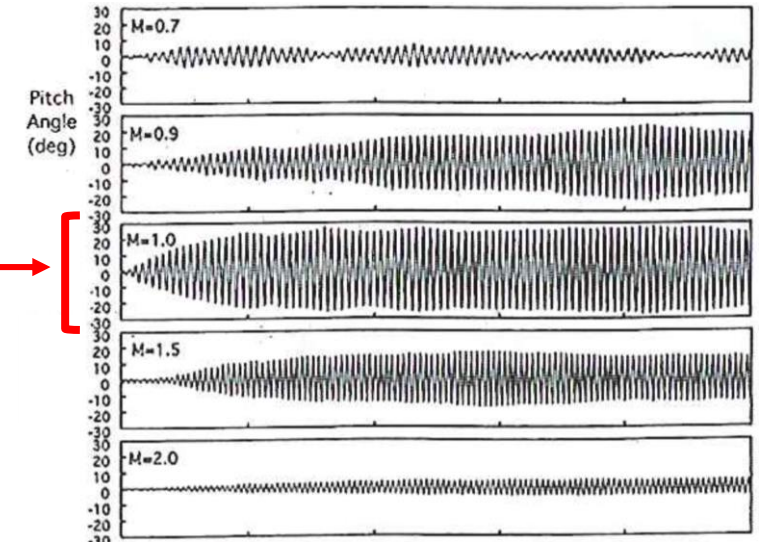


**Hayabusa Capsule Recovery
in Australia**

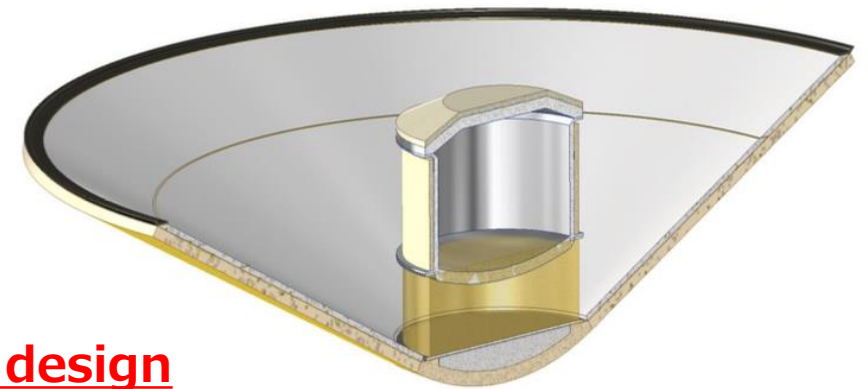
Introduction



- Dynamic Instability in the Transonic Regime [1]
 - Landing prediction accuracy reduced
 - Introduces risk of parachute deployment failure
- Deep-Space Capsule (DS-SRC) Design Requirements [2]
 - Higher re-entry speeds (16 kms^{-1})
 - Lower mass (10 kg)



New Aeroshell Design Proposed



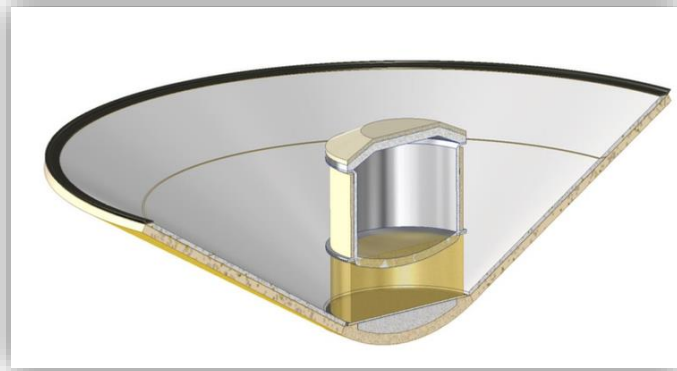
$\therefore$ It is necessary to study the dynamic behaviour of this new design

[1] Hiraki, K., "Experimental Study on Dynamic Instability of Capsule-shaped Body," ISAS report, 103,, pp1-55, 1999.

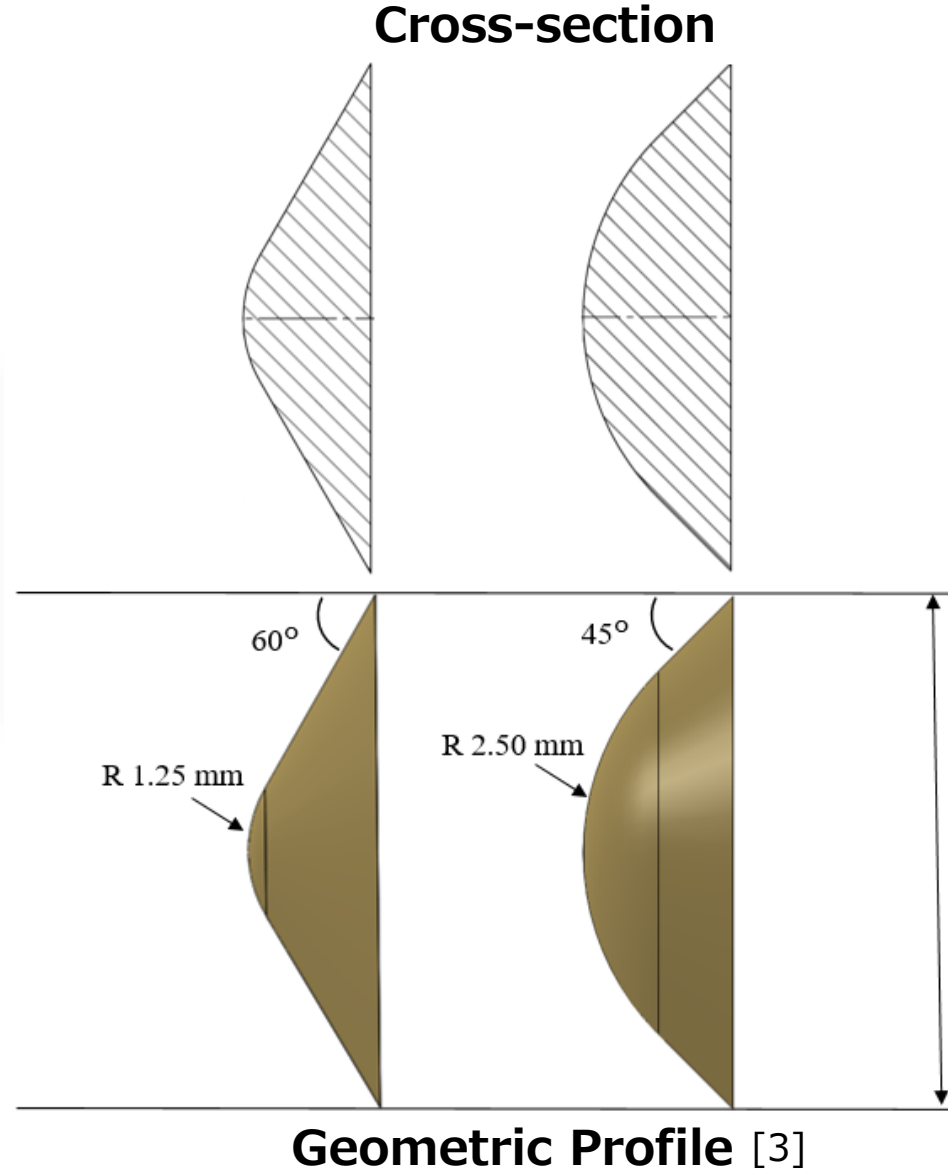
[2] Yamada, K., "Research and Development of Sample Return Capsule for Future Deep Space Exploration," Symposium on Shock Waves., 2019.

Experimental Setup

■ Brass Model Design



Reference DS-SRC Model



Hayabusa-like DS-SRC Model

[2] Yamada, K., "Research and Development of Sample Return Capsule for Future Deep Space Exploration," Symposium on Shock Waves., 2019.

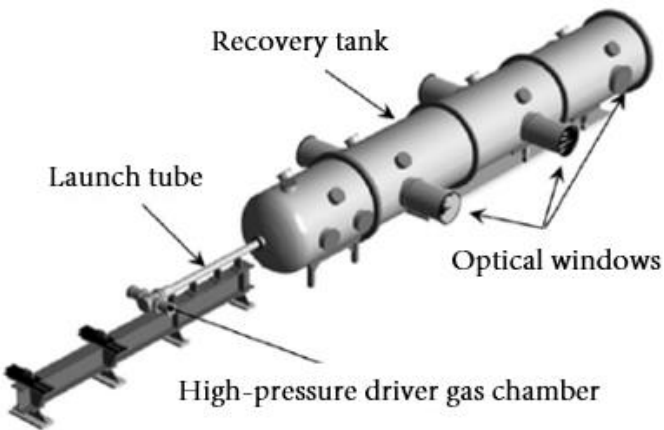
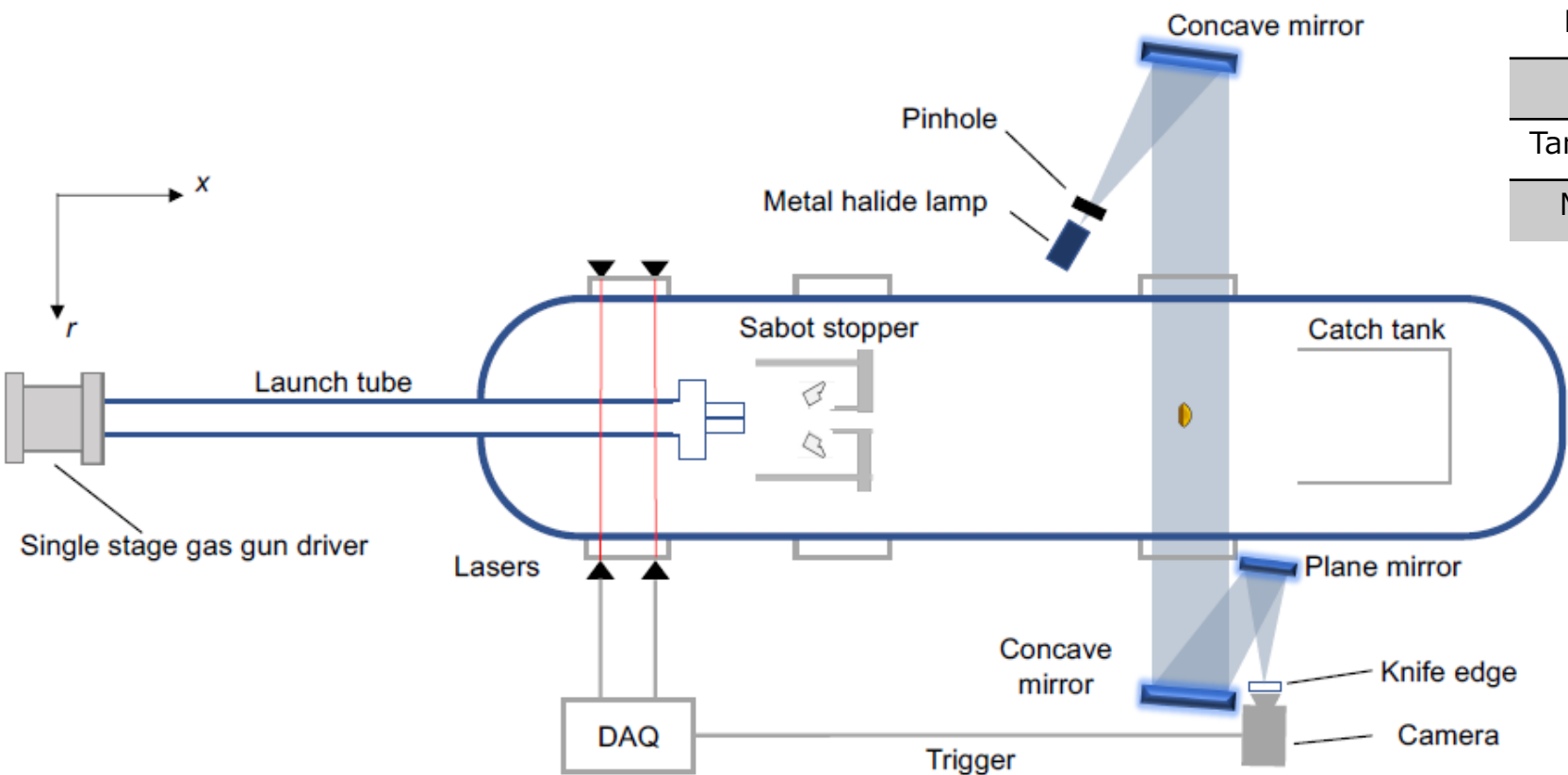
[3] Takasawa, H., "Static Attitude Stability of Deep Space Sample Return Capsule with Thin Aeroshell" CEAS Space Journal., 2024.

Experimental Setup



■ Ballistic Range and Schlieren Imaging [4]

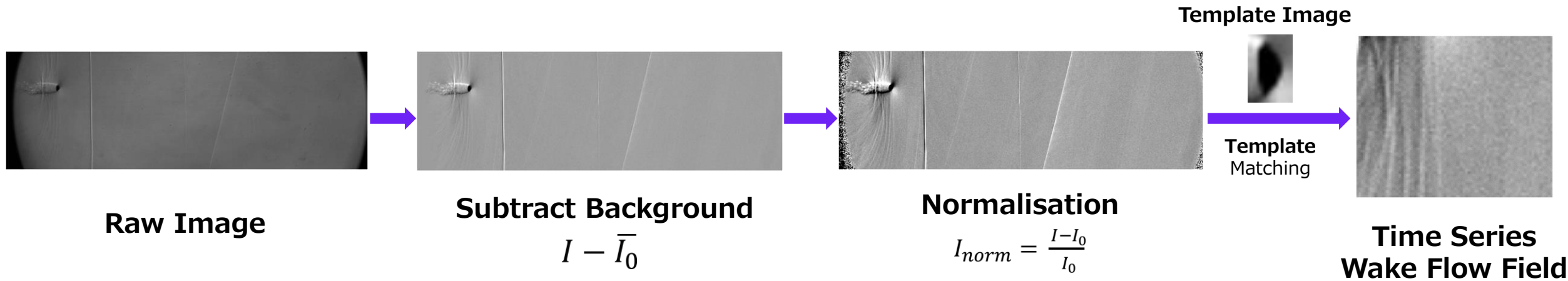
Parameter	Value
Frame Rate [fps]	130,000
Exposure Time [μs]	0.46
Resolution [px]	640 x 208
Target Mach Number [-]	0.8 – 1.2 (transonic)
Model Diameter [m]	0.0048



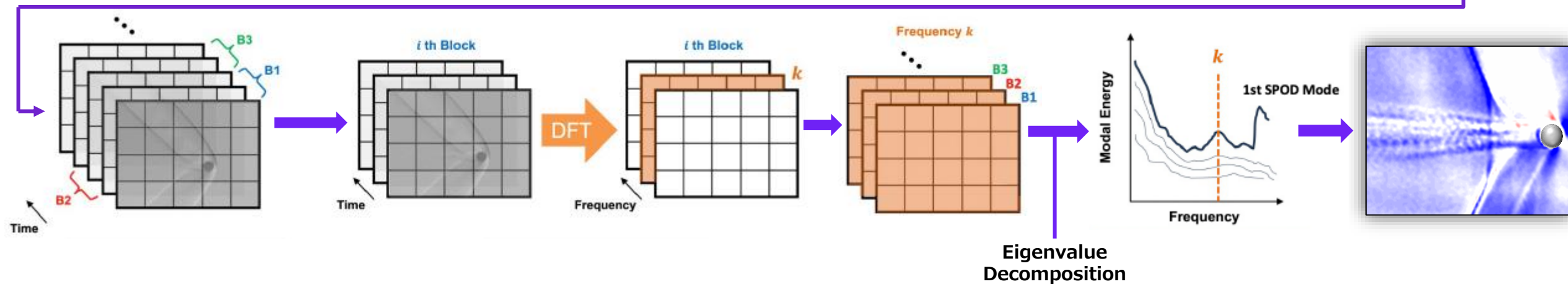
[4] Kim, J., "Experimental Investigation of Afterbody Geometry Effect on Dynamic Instability of Atmospheric Entry Capsule in Transonic Regime," ICFD2025, 2025.

Post-Processing Techniques

■ Raw Image Processing [4]:



■ Spectral Proper Orthogonal Decomposition (SPOD):



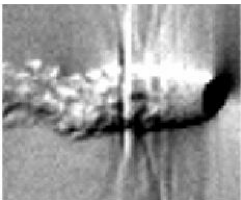
- Summary of Experimental Campaign and Raw Data

	5mm Sphere	Hayabusa	Reference	Total
Num. Trials	5	7	4	16
Success	1	1	0	3
Visible, Unusable	1	3	1	4
Not Visible	3	3	3	9

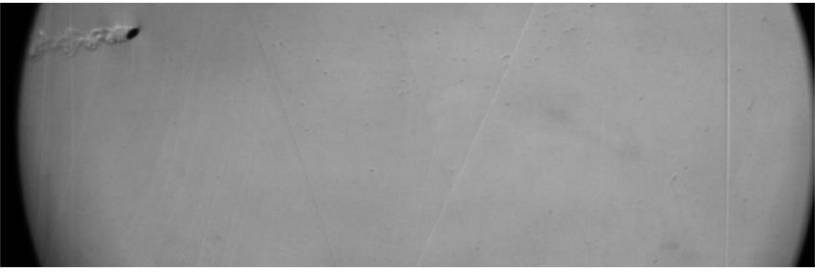
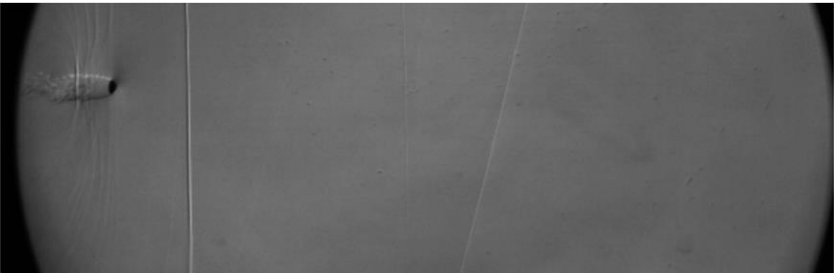
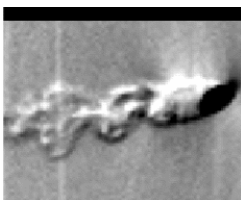
5 mm Sphere Model - Mach 1.02



Hayabusa-like Model - Mach 0.89



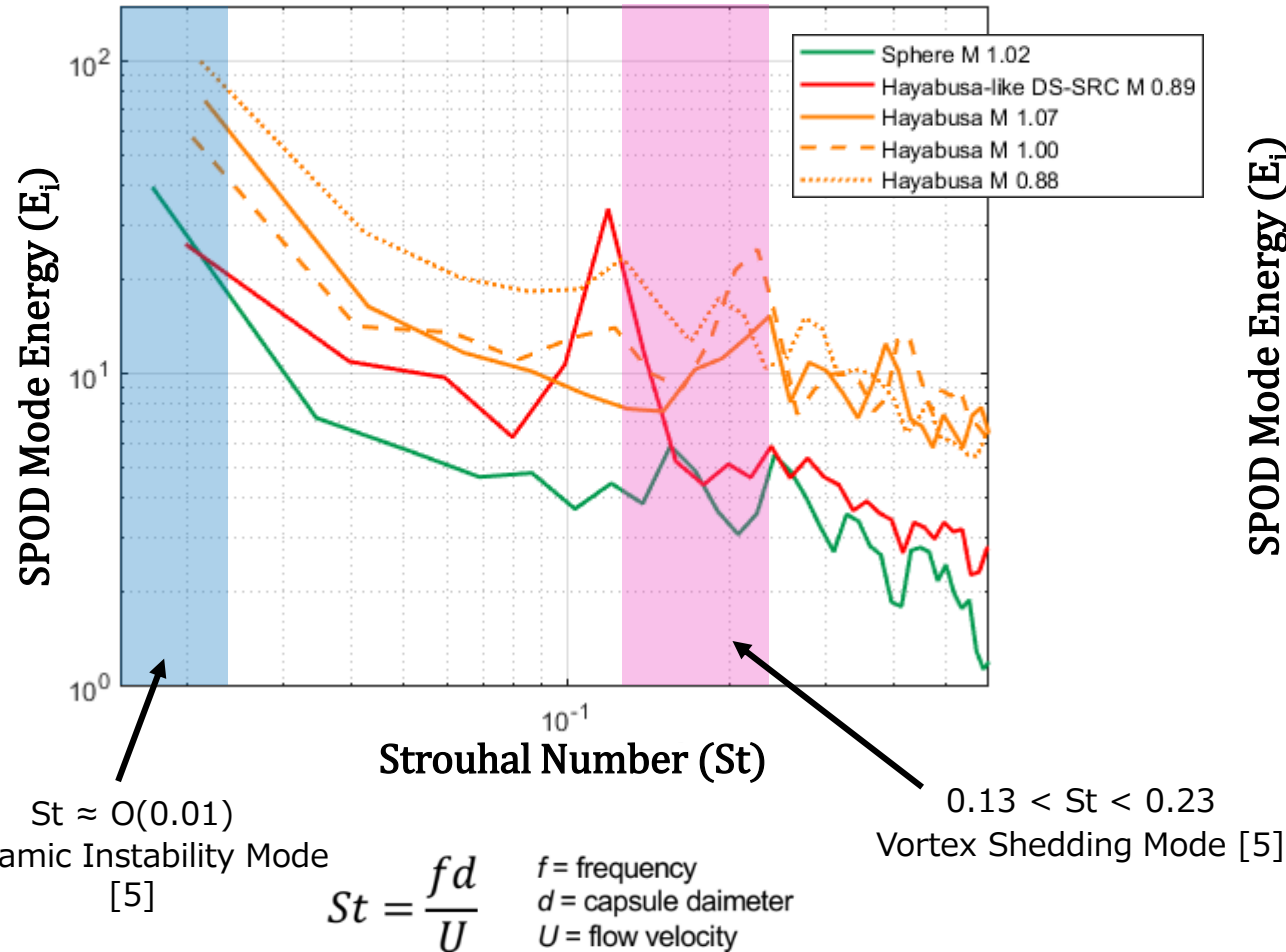
Reference Model - Mach 0.78 (unusable)



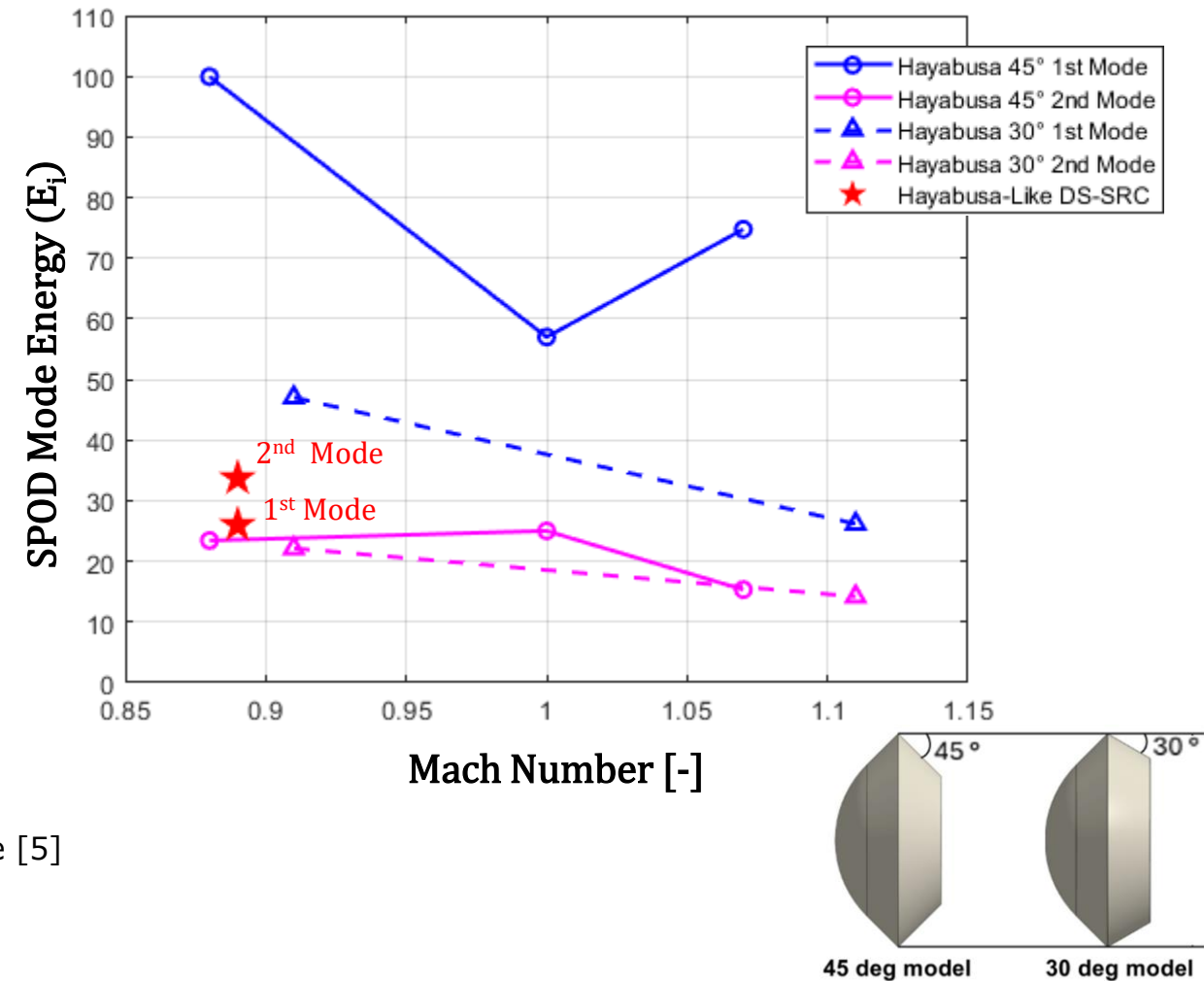
Results

■ SPOD Mode Energy Analysis

SPOD Mode Energy vs Strouhal Number



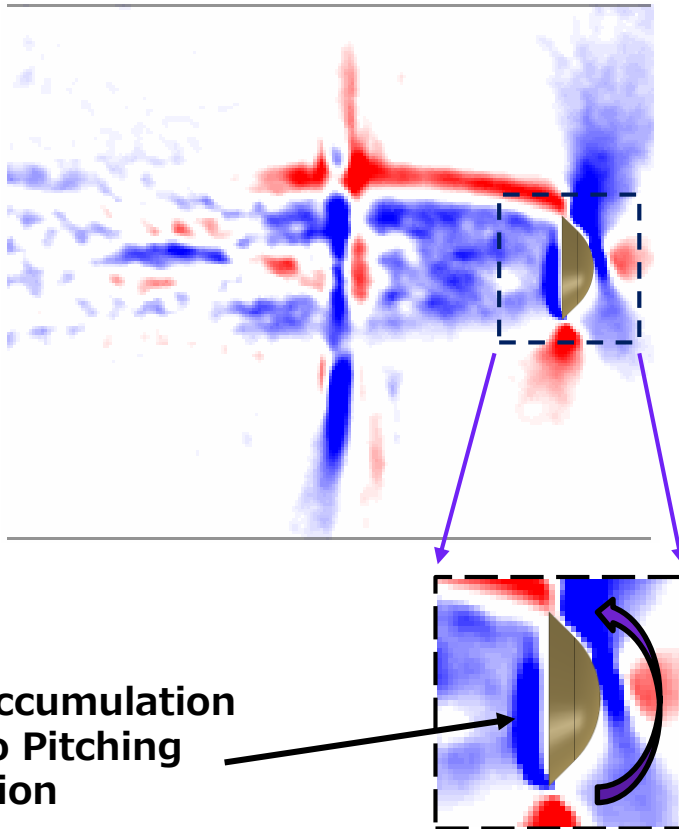
SPOD Mode Energy vs Mach Number



■ SPOD Analysis

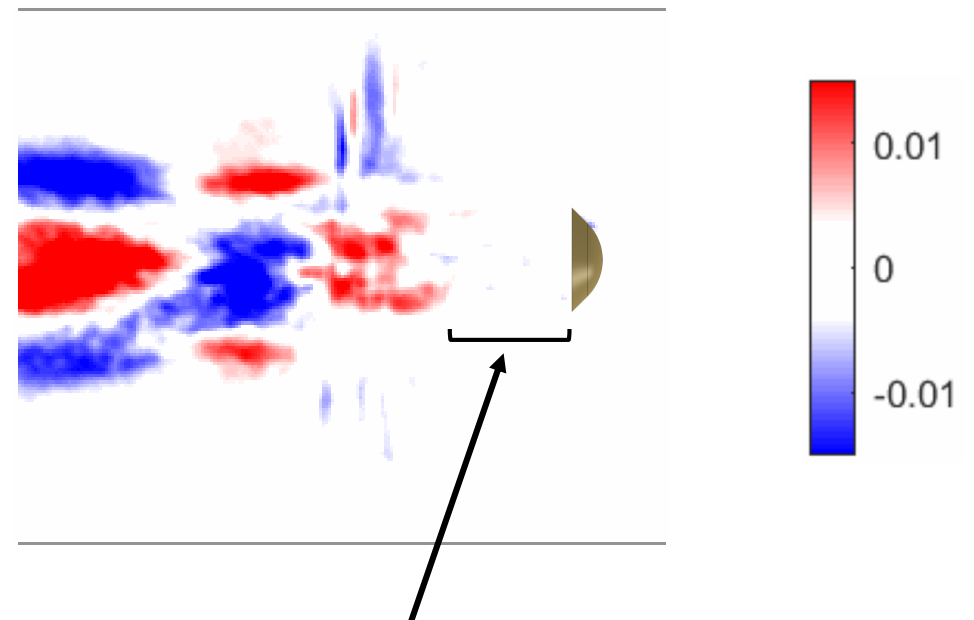
- Dynamic Instability: $St \approx O(0.01)$, and vortex shedding at $0.13 < St < 0.23$ [5]

$St = 0.02$



Back Pressure Accumulation
Contributes to Pitching
Oscillation

$St = 0.12$



No impact of vortex
shedding observed near the
capsule

Conclusion

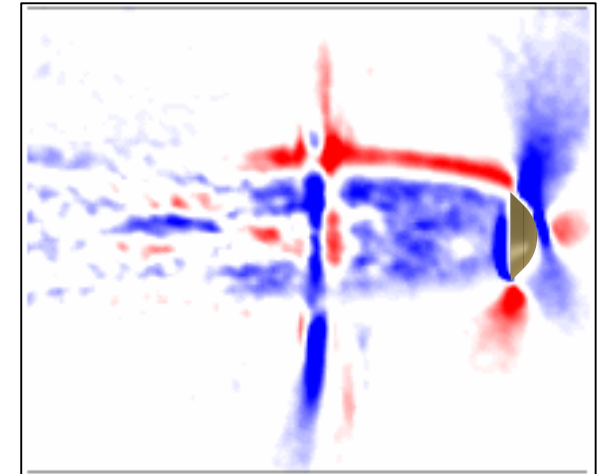
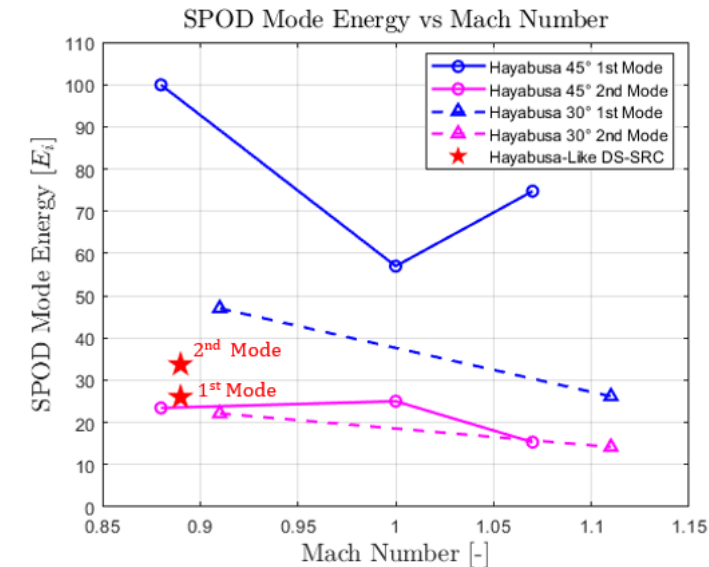


Key Outcomes

- Spatio-temporal analysis performed on a novel Hayabusa-like DS-SRC design
- Dynamic instability observed for a Hayabusa-like DS-SRC
 - Link formed with back pressure oscillation
 - Mode dominance compared with previous studies
- Foundation and proof of concept for a much deeper investigation

Future Avenues of Investigation

- Free-flight investigation into dynamic instability of JAXA's proposed next-generation DS-SRC
 - Compare with Hayabusa-like DS-SRC to investigate impact of forebody geometry
- Data points at higher Mach numbers
 - Decrease tank pressure
 - Use an accelerating gas with a lower molecular weight (e.g. helium)



Thank You!

Questions?